

Department of Computer Sciences
CS-1005 – Discrete Structures
Semester: Spring 2026

Assignment #2

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Due Date: As mentioned on Google Classroom
Max Score: 322 Marks

Note:

- Please submit handwritten solutions to all of the questions given below.
- Scan your handwritten solution and upload to Google Classroom.
- No extension in deadline.
- Plagiarism will result in zero marks.

Q1 – Basic Proofs [30 Marks]

(a) [5 Marks] — Prove that for any odd integer n , the following equation holds:

$$\lfloor n^2 / 4 \rfloor = (n^2 - 1) / 4$$

(b) [12 Marks] — Prove that for all integers $n \geq 1$, the expression $7^n - 2^n$ is divisible by 5.

You must prove this using **two distinct methods**:

1. **Method 1:** A direct proof using algebraic factorization. [6 Marks]
2. **Method 2:** Mathematical Induction. [6 Marks]

(c) [13 Marks] — Consider the sequence s_n defined recursively as follows:

$$s_1 = 3$$

$$s_2 = 18$$

$$s_k = 6 s_{k-1} - 9 s_{k-2} \quad \text{for all integers } k \geq 3$$

3. **Part 1 [8 Marks]:** Prove by **Strong Mathematical Induction** that for all integers $n \geq 1$, the explicit formula for this sequence is:

$$s_n = n \cdot 3^n$$

4. **Part 2 [5 Marks]:** Using the explicit formula from Part 1, prove the following *biconditional* statement:

“The term s_n is an even integer *if and only if* n is an even integer.”

Q2 – Recurrence Relations [20 Marks]

Part 1 [2 + 6 + 2 = 10 Marks] — You are a cybersecurity analyst tracking a rogue AI virus known as “Project Hydra.” You’ve noticed a terrifying pattern in how it infiltrates secure network nodes.

At hour 0, exactly 1 node was compromised.

At hour 1, exactly 4 nodes were compromised.

Every hour thereafter, the AI mutates based on its previous states: the number of nodes compromised is a difference of exactly 3 times the previous hour’s compromised nodes, *and* 2 times the nodes compromised two hours ago (because your automated firewall detects and quarantines the oldest infections).

The system will permanently crash if **3,000 or more nodes** are infected in a single hour.

The Task:

5. **[2 Marks]** Write the recurrence relation and initial conditions for the sequence of infected nodes a_n .
6. **[6 Marks]** Solve the recurrence relation to find the explicit (closed-form) formula for a_n .
7. **[2 Marks]** Exactly how many hours (n) do you have before the system crashes?

Part 2 [5 Marks] — You have intercepted a heavily encrypted hard drive. The drive’s defense mechanism is a rotating quantum passcode that changes every minute k . The machine generates the next key based on this rule:

$$c_k = c_{k-1} / (3 c_{k-1} + 1)$$

for every integer $k \geq 2$, with initial key $c_1 = 1$.

If you try to unlock the drive with the wrong key, it wipes the data. Discover the **exact explicit formula** for c_k using the **forward iterative method**.

Part 3 [2 + 1 + 2 = 5 Marks] — A new ride-sharing app uses an aggressive referral algorithm. The company starts by giving premium accounts to 4 local influencers (Generation 1). Each influencer recruits exactly 6 new users to keep their premium status (Generation 2), and the strict 1-to-6 recruitment tree continues.

8. **[2 Marks]** How many **new** users are recruited during exactly the 7th generation?
9. **[1 Mark]** Write the explicit formula for the number of new users recruited in Generation n .
10. **[2 Marks]** What is the closed-form sum formula for the **total** number of users (from all generations combined) who have joined the app by Generation n ?

Q3 – TV Multiverse [55 Marks]

Clearly state your assumptions and show all steps for mathematical proofs.

1. The Laundering Sequence [5 Marks]

Saul Goodman tracks his “legal fees” using an arithmetic sequence where the n -th term is:

$$a_n = 10 + (n - 1) \cdot 4$$

Calculate the total sum of the first **50** entries in his ledger.

2. The Escape Route Logic [12 Marks]

Michael Scofield finds an encrypted message:

“For every integer k , if k^2 is even, then k is even.”

To verify this for the escape plan, provide **two distinct proofs**:

- Prove it using **Contraposition**. [6 Marks]
- Prove it using **Contradiction**. [6 Marks]

3. The Export Tax [5 Marks]

The Shelby Company Ltd. exports gin in barrels that must weigh exactly **80 kg** for customs. If a batch weighs **W** kg, express the number of **full** barrels that can be exported using floor or ceiling notation. Which notation is mathematically correct to ensure no barrel is underweight?

4. The Dark Army Ledger [10 Marks]

The Dark Army’s recruitment grows by the rule:

$$s_k = s_{k-1} + 2k$$

Use the **Method of Iteration** to find the closed-form solution for this recruitment rate, assuming $s_0 = 0$.

5. The Siege of Paris [12 Marks]

Ragnar states that a successful raid requires specific troop divisions. Prove the following:

“For all integers n , p , and q , if n divides p and n divides q , then n must also divide the linear combination **(2p – 3q)**.”

6. The Limbo Recurrence [11 Marks]

A deep-level dream sequence follows the recurrence:

$$a_0 = 3, \quad a_1 = 13, \quad a_k = 6 a_{k-1} - 5 a_{k-2}$$

Use **Strong Induction** to prove that the explicit formula is:

$$a_n = (5/2) \cdot 5^n + 1/2$$

Q4 – Abdul Bari’s Legacy [12 Marks]

Part A) The Gate of Good Deeds

In the village, Abdul Bari and his friend are learning about the balance of good deeds. Abdul Bari carries a bag of dates representing an even integer r . His friend carries a bag representing an odd integer s .

They arrive at a beautiful garden gate. The gatekeeper tells them: “To enter, you must prove the property of your bags. Take three times Abdul Bari’s dates and add it to twice your friend’s dates. Prove to me that this total will always result in an even number of dates.”

Your task is to help Abdul Bari and his friend:

11. **[2 Marks] Extract the Statement:** Identify the specific mathematical claim that needs to be proven.
12. **[5 Marks] Prove the Statement:** Use a clear, formal proof using the definitions of even and odd integers.

Part B) The Perfect Square Charity [5 Marks]

Abdul Bari wants to distribute coins to the needy in an organized way. He makes a claim:

“Every positive even integer less than 26 can be expressed as a sum of three or fewer perfect squares.”

Prove this statement by the **Method of Exhaustion**.

(Note: A perfect square is a number like 1, 4, 9, 16, ...)

Q5 – The Vercetti Estate Syndicate Operations [35 Marks]

Tommy Vercetti has just consolidated power in Vice City, but to keep the empire running, he needs a crew that can handle more than just an M4. The previous owners left a series of encrypted ledgers and structural puzzles. To prove you’re worth more than a common street thug, solve all parts of this operational challenge:

Part A) Parity of Product [10 Marks]

Avery Carrington needs to verify the **Parity of Product** for the construction of the Star Island mansion. Prove that for any two integers m and n :

- If the product $m \cdot n$ is **even**, then at least one of m or n must be **even**.

Furthermore, prove the **Quotient-Remainder Theorem** for $d = 2$: for every integer n , there exists a unique integer q such that $n = 2q$ or $n = 2q + 1$.

Part B) Lance Vance's Delivery Sequences [6 Marks]

Lance Vance is tracking “deliveries” made across the city. He has recorded the following sequences and needs you to find their **explicit formulas** so he can predict future shipments. [2 Marks each]

11. 2, 7, 12, 17, 22, ...

12. 1, 8, 27, 64, 125, ...

13. $\frac{1}{3}$, $\frac{2}{4}$, $\frac{3}{5}$, $\frac{4}{6}$, ...

Part C) Induction: Sum of Cubes [10 Marks]

The security vault at the El Banco Corrupto Grande Bank in Little Havana uses a summation-based lock. Use **Mathematical Induction** to prove that for all integers $n \geq 1$:

$$1^3 + 2^3 + 3^3 + \dots + n^3 = n^2(n+1)^2 / 4$$

Part D) Induction: Divisibility [9 Marks]

A mechanical glitch follows a divisibility pattern. Use **Mathematical Induction** to prove that for every integer $n \geq 1$:

$$7^n - 1 \text{ is divisible by } 6$$

Q6 – The Archivist's Notebook [100 Marks]

Somewhere in the basement of a decommissioned records office, a notebook was found. It contained four procedures written out in careful handwriting, and a single line written across the inside cover:

“Every document has one address. Every address holds one document. The address can always be recovered.”

No context. No explanation. Just the four procedures, the claim, and the blank page that follows.

Your job is to determine whether the claim is true. You cannot cite any result by name. You cannot appeal to anything from outside this course. Every lemma you need must be built from scratch using only the definitions given here and the proof techniques you have studied. Any step that is not proved is not credited. The six statements listed after the code are conjectures. Do not assume any of them until you have proved them yourself.

Read all four procedures carefully before attempting any part of this assignment.

The Procedures

```
// isqrt(m): returns the largest integer r >= 0 such that r * r <= m
int isqrt(int m) {
    int r = 0;
    while ((r + 1) * (r + 1) <= m)
```

```

        r = r + 1;
    return r;
}

// -----
// k, x, s, tag are output parameters written by the procedure.
void STAMP(int n, int k, int x, int s, int tag) {
    int a = 1, b = 0;
    k = 0; s = 0; tag = 0;\
    int pwr = 1;
    while (a + b <= n) {
        a = a + 2 * b;
        b = a - b;
        s = s + b;
        tag = tag + pwr * (b % 2);
        pwr = pwr * 2;
        k = k + 1;
    }
    x = n - b;
}

// -----
int RESTORE(int k, int x) {
    int a = 1, b = 0, i = 0;
    while (i < k) {
        a = a + 2 * b;
        b = a - b;
        i = i + 1;
    }
    return b + x;
}

// -----
int DECODE(int k, int x, int s) {
    int c = 2 * s + 1;
    int D;
    if (k % 2 == 0)
        D = 2 * c * c + 2;
    else
        D = 2 * c * c - 2;
    int r = isqrt(D);
    int b = (c - (r - c)) / 2;
    return b + x;
}

// -----
bool VERIFY(int k, int x, int s, int tag) {
    int p = 1, q = 0, j = 1;
    while (j <= k) {
        p = p + 2 * q;
        q = p - q;
        j = j + 1;
    }
    bool check_A = (p*p - 2*q*q == 1) || (p*p - 2*q*q == -1);
    bool check_B = (x >= 0) && (x < p);
    bool check_C = (2*s + 1 == p + 2*q);
    return check_A && check_B && check_C;
}

```

The Six Claims

The following six statements are what you are ultimately working toward. You may **not** use any of them until you have proved them.

- (i) STAMP terminates on every positive integer input.
- (ii) $\text{RESTORE}(k, x) = n$ whenever $(k, x, s, \text{tag}) = \text{STAMP}(n)$.
- (iii) $\text{DECODE}(k, x, s) = n$ whenever $(k, x, s, \text{tag}) = \text{STAMP}(n)$.
- (iv) $\text{VERIFY}(k, x, s, \text{tag})$ returns true whenever $(k, x, s, \text{tag}) = \text{STAMP}(n)$.
- (v) If $\text{STAMP}(m) = \text{STAMP}(n)$ then $m = n$.
- (vi) Every four-tuple for which VERIFY returns true is produced by $\text{STAMP}(n)$ for exactly one positive integer n .

Part I — The Engine [20 Marks]

All four procedures contain loops that apply the same two-line update to the same two variables. Before anything else, you need to understand what that update does as a mathematical object.

[4 Marks] **(a)** Look at any of the four loops. The two-line update applied to a and b appears in all of them unchanged. Extract it and write it as a pair of recurrences. Define sequences $A(0), A(1), A(2), \dots$ and $B(0), B(1), B(2), \dots$ using the initial values of a and b and the update rule. State both recurrences and both initial conditions precisely.

Warning. The two assignments inside the loop body are *sequential*, not simultaneous. The variable updated on the first line holds its new value when the second line executes. If you do not account for this, your recurrences will be wrong, and everything downstream will be wrong with them.

[6 Marks] **(b)** Derive closed-form expressions for $A(n)$ and $B(n)$. Assume a solution of the form r^n , substitute into your recurrence, factor the resulting equation, find both roots, write the general solution as a linear combination, and determine the free constants from the initial conditions. Show every algebraic step. The derivation itself is graded, not just the final expressions.

[7 Marks] **(c)** Prove each of the following four claims using only the recurrences from part (a). The closed forms from part (b) may **not** be used here. Each claim needs its own proof.

- (i) $A(n) \geq 1$ for all $n \geq 0$.
- (ii) $B(n) \geq 0$ for all $n \geq 0$, with equality only when $n = 0$.
- (iii) $A(n)$ is strictly increasing for $n \geq 1$.
- (iv) $B(n)$ is strictly increasing for $n \geq 1$.

[3 Marks] **(d)** Prove that $A(n)$ is odd for every $n \geq 0$.

Part II — The Identity [14 Marks]

[7 Marks] **(a)** Compute $A(n)^2 - 2 \cdot B(n)^2$ for $n = 0, 1, 2, 3, 4, 5$. Write down the pattern you observe as a precise conjecture about the value of this expression for all $n \geq 0$. Then prove the conjecture holds for all n .

Your proof must be written out in full. Every algebraic step in the inductive step must appear on the page without skipping. Before writing the proof, state explicitly whether weak induction is sufficient or not, and explain why.

[7 Marks] **(b)** Prove that there is no pair of integers p, q with $q > 0$ satisfying $p^2 = 2 \cdot q^2$.

This proof has a constraint that is strictly enforced. You must use the identity from Part II(a) as the central tool of the argument. The following are **forbidden**:

- Any argument that examines whether p or q is even or odd.
- Any argument involving divisibility by 2 or prime factorization.
- Any form of the standard parity argument in any disguise.

The only arithmetic facts you may use freely are that the integers are closed under addition, subtraction, and multiplication, and that a nonzero integer has absolute value at least 1. Every other step must follow from Part II(a) and the properties of A and B established in Part I.

Part III — The Filing Machine [20 Marks]

[5 Marks] **(a)** Prove that STAMP terminates on every positive integer input n . You must identify an explicit integer-valued expression involving the loop variables, prove it is non-negative throughout the loop, and prove it decreases strictly with each iteration. The expression must be stated clearly before the proof begins.

[7 Marks] **(b)** State a single loop invariant for STAMP that is strong enough to imply all five of the following conditions at the moment the loop exits:

- (i) $a = A(k)$
- (ii) $b = B(k)$
- (iii) $B(k) \leq n$ and $A(k) + B(k) > n$
- (iv) $s = B(1) + B(2) + \dots + B(k)$
- (v) $\text{pwr} = 2^k$

Write four separate proofs: that the invariant holds before the loop begins; that one arbitrary iteration preserves it; that on exit the invariant implies condition (iii); and that on exit it implies conditions (i), (ii), (iv), and (v). A single blended argument will not be accepted.

[3 Marks] **(c)** Using only the invariant and exit conditions from part (b), prove that the return value $x = n - b$ satisfies $0 \leq x < A(k)$. State precisely which exit conditions you are using and at which point in the argument.

[5 Marks] **(d)** Prove the identity: $s = (A(k+1) - 1) / 2$. First prove that $A(k+1) - 1$ is always even, using Part I(d) and the recurrence for A . Then prove the sum identity by induction. You will need both recurrences at the same time in the inductive step.

Part IV — The Recovery [8 Marks]

[4 Marks] **(a)** Write a loop invariant for RESTORE that captures the relationship between the variables a , b , i and the sequences A and B at the start of each iteration.

[4 Marks] **(b)** Prove Claim (ii): $\text{RESTORE}(k, x) = n$ whenever $(k, x, s, \text{tag}) = \text{STAMP}(n)$. Your proof must invoke the invariant from part (a) by name at the point where it is applied. It must also invoke exit condition III(b)(iii) by name at the point where it is used.

Part V — The Shortcut [12 Marks]

[2 Marks] **(a)** Using Part III(d), identify exactly what the quantity $c = 2 \cdot s + 1$ represents in terms of the sequences A and B . State it as an equation and prove it.

[4 Marks] **(b)** Prove that the value D computed by DECODE is always a perfect square. Handle the cases k even and k odd separately. In each case, state which value D is the square of and derive it using Part II(a) as a lemma. Do not guess the square root and verify it afterwards. Show every algebraic step.

[2 Marks] **(c)** Prove as a direct corollary of part (b) that $\text{isqrt}(D)$ returns a value r satisfying $r^2 = D$ exactly, with no rounding. State this explicitly as a corollary and prove it from part (b).

[3 Marks] **(d)** Prove that the expression $(c - (r - c)) / 2$ computed inside DECODE equals $B(k)$. First simplify the expression algebraically. Then prove the operand of the division by 2 is always even.

[1 Mark] **(e)** Combining parts (a) through (d), prove Claim (iii): $\text{DECODE}(k, x, s) = n$ whenever $(k, x, s, \text{tag}) = \text{STAMP}(n)$. List the results you are invoking and the order in which you use them.

Part VI — The Inspector [16 Marks]

[4 Marks] **(a)** The loop in VERIFY has a different structure from the loops in STAMP and RESTORE. Despite this, prove that after k iterations, the variables p and q in VERIFY hold the same values that a and b hold in STAMP and RESTORE after k iterations.

[6 Marks] **(b)** Prove Claim (iv): $\text{VERIFY}(k, x, s, \text{tag})$ returns true whenever $(k, x, s, \text{tag}) = \text{STAMP}(n)$. Write three separate sub-proofs, one for check_A , one for check_B , and one for check_C . Each sub-proof must name the earlier result it depends on. Merged arguments will not be accepted.

[6 Marks] **(c)** VERIFY computes tag during its loop but does not use it in any of the three checks. Determine what value tag holds on exit from STAMP for a given k . Work through the following steps in order:

- Prove by induction that $B(j)$ is odd if and only if j is odd.
- Use this to determine exactly what value is added to tag on each iteration of STAMP's loop.
- Sum these contributions and derive a formula for tag as a function of k alone.
- Simplify the formula as far as you can.
- Verify your formula by tracing STAMP manually for $k = 1, 2, 3, 4, 5$.

Then answer the following question with a proof: given only k and no other information, can tag always be recovered exactly?

Part VII — The Claim [10 Marks]

[4 Marks] **(a)** Prove Claim (v): if $\text{STAMP}(m) = \text{STAMP}(n)$ then $m = n$. Do not prove this by applying RESTORE to both m and n and citing Claim (ii). That argument is circular at this point in the assignment. Instead, work directly from the loop invariant and exit conditions of STAMP.

[6 Marks] **(b)** Prove Claim (vi) in two steps:

Step 1 (Existence). Prove that for every pair of non-negative integers (k, x) with $k \geq 1$ and $0 \leq x < A(k)$, there exists a positive integer n such that $\text{STAMP}(n)$ returns first component k and second component x . You must construct n explicitly and prove the construction is correct. The proof must use strong induction on k .

Step 2 (Completeness). Prove that if $\text{VERIFY}(k, x, s, \text{tag})$ returns true, then the pair (k, x) satisfies the hypotheses of Step 1, and the value of s in the four-tuple matches what STAMP would produce for the integer n constructed in Step 1. From this, conclude formally that Claim (vi) holds.

Q7 – The ORIN Protocol [50 Marks]

Somewhere in an encrypted forum whose address changes every seventy-two hours, a collective known as the Athenaeum published a protocol. They called it ORIN: the Onion Routing Inspired Network. A message passes through a chain of k relay stations. Each station removes exactly one layer and forwards the rest. When the last station removes its layer, the original message is recovered.

“Every layer added must be a layer that can be removed. The key ladder guarantees this. Everything else follows from the key ladder.” — *The Athenaeum*

You may not cite any result by name. Everything must be derived from the definitions below.

The ORIN Definitions

Definition 1 (The Key Ladder). Fix a seed pair $(K(0), L(0)) = (1, 0)$. For each $n \geq 1$ define:

$$K(n) = K(n-1) + 2 \cdot L(n-1)$$

$$L(n) = K(n-1) + L(n-1)$$

The key assigned to relay station j is $K(j)$. The sequence $K(1), K(2), \dots, K(k)$ is the key ladder of a circuit of depth k .

Definition 2 (Layering and Relay Selection). A k -layer packet is built from a payload M of s bytes as follows. Set $P_0 = M$. For $j = 1, 2, \dots, k$ set $P_j = \text{LAYER}(P_{j-1}, K(j))$, where each application of LAYER increases the packet size by exactly 512 bytes. The function STRIP is the exact inverse of LAYER. Relay selection: station i is selected by selector value v when $W(1) + \dots + W(i-1) \leq v < W(1) + \dots + W(i)$.

Definition 3 (Cell Transmission). Data is transmitted in fixed-size cells of 512 bytes. A packet of m bytes occupies $\lceil m / 512 \rceil$ cells, with the final cell padded to 512 bytes if necessary.

Part I — The Key Ladder [23 Marks]

[4 Marks] **(a)** The two recurrences in Definition 1 are coupled. Eliminate L from the system to derive a single second-order recurrence for $K(n)$ alone. Show every substitution step. Then state the corresponding recurrence for $L(n)$ alone.

[8 Marks] **(b)** Prove each of the following four claims using only the recurrences from part (a). The closed forms may not be used here. Each requires its own proof.

- (i) $K(n) \geq 1$ for all $n \geq 0$.
- (ii) $L(n) \geq 1$ for all $n \geq 1$.
- (iii) $K(n)$ is strictly increasing for $n \geq 1$.
- (iv) $K(n)$ is odd for all $n \geq 0$.

[6 Marks] **(c)** Compute $K(n)^2 - 2 \cdot L(n)^2$ for $n = 0, 1, 2, 3, 4$. State the pattern as a precise conjecture. Before writing the proof, state whether weak induction is the right form and explain why. Then prove the conjecture holds for all $n \geq 0$, writing every algebraic step in the inductive case without omitting any.

[5 Marks] **(d)** Prove that $L(1) + L(2) + \dots + L(k) = (K(k+1) - 1) / 2$ for all $k \geq 0$. First prove $K(k+1) - 1$ is always even using part (b)(iv). Then prove the sum identity by induction. The inductive step requires both recurrences at the same time.

Part II — Layering and Unlayering [16 Marks]

[4 Marks] **(a)** Let $S(k)$ denote the byte size of a k -layer packet built from a payload of s bytes. Write a recurrence for $S(k)$ using Definition 2 and solve it. Prove by induction that $S(k) = s + 512 \cdot k$ for all $k \geq 0$.

[6 Marks] **(b)** The layering process builds $P_0, P_1, \dots, P_k$ by applying LAYER at each step. State a condition that holds after exactly j steps of this process. Prove this condition holds before the process begins, that each step preserves it, and that after k steps it implies the result is a valid k -layer packet keyed by $K(1)$ through $K(k)$ in order.

[6 Marks] **(c)** The unlayering process starts from P_k and applies STRIP at steps $j = k, k-1, \dots, 1$. State a condition that holds after the process has completed steps $k, k-1, \dots, k-i+1$. Prove this condition holds at the start, is preserved at each step, and implies on exit that the result equals M . At the point where you use the condition from part (b), say so explicitly.

Part III — Relay Selection and Cell Transmission [11 Marks]

[5 Marks] **(a)** Suppose all n relay stations have equal weight W , so $T = n \cdot W$. Using the quotient-remainder theorem, prove that the station selected by any $v \in [0, nW-1]$ is $(v \text{ div } W) + 1$. Then prove by contradiction that for $n \geq 2$, no station can be selected by exactly T/n^2 values of v when n does not divide W .

[6 Marks] **(b)** Prove that the padding added to the final cell of any packet of $m \geq 1$ bytes lies in the range $[0, 511]$. Prove the bound is tight by constructing an explicit m for which the padding is exactly 511, and prove no m gives padding larger than 511. Then prove that a k -layer packet built from a payload of s bytes occupies exactly $\lceil s / 512 \rceil + k$ cells. Derive this by first proving the identity $\lceil (m + 512 \cdot k) / 512 \rceil = \lceil m / 512 \rceil + k$ for all $m \geq 1$ and $k \geq 0$.

Q8 – The Accountant of Nothing [20 Marks]

A question on Mathematical Induction in three descents

“The Pit does not ask why you fell. It already knows. It just does not care.”

Nobody remembers who built the Pit. This is fitting, because the Pit does not remember either. Memory requires a floor to stand on. Something solid below you, some point where things stop. A base case, if you will.

The Pit has no base case.

What it does have is structure. That is the part that gets you. Not the darkness, not the cold, not the falling itself. It is the structure. Infinite and careful and completely without purpose, like a cathedral built in a language nobody speaks, for a god that never showed up.

A man named **Soren** fell into the Pit on a Tuesday. He did not trip. He stepped in on purpose, after finishing a cup of coffee that was, in his words, fine. Just fine. He had spent seventeen years doing careful philosophy and had arrived at the conclusion that continuing was a choice, and this particular Tuesday he chose otherwise. He did not leave a note. There was nothing he had not already written down, and none of it had helped.

He has been falling ever since. He does not age. He does not sleep. He does not stop being a mathematician. He has come to regard this last part as a kind of cosmic punchline. Not funny, exactly. But a punchline.

(a) The Grey Stones [5 Marks]

At each level of the Pit, Soren passes a collection of grey stones. They are not gravestones. There is no one buried here, and the word grave implies something ended, which implies something once lived with intention. These stones have never had intentions. They were simply placed, at some point, by something that did not explain itself, and they have been here since.

At level k , there are exactly k stones. One at the first level. Two more at the second. Three at the third. The Pit adds k stones at level k , as though it is keeping score of something it will never announce. A

ledger. No purpose, no reader, no point. Just the numbers, accumulating, the way certain things accumulate whether you want them to or not.

Soren counts them. Not because counting helps. Not because he expects the total to mean anything. He counts because he is a mathematician and that particular habit does not leave you, even here, even now, even after everything.

He says the total number of stones through level n is:

$$T(n) = 1 + 2 + 3 + \dots + n = n(n+1) / 2$$

Prove this by mathematical induction.

“The stones have no comment on this. The stones have never had a comment on anything.”

(b) The Words for Nothing [6 Marks]

The stones are not blank.

Of course they are not blank. The Pit is not a careless place. If it were careless, it might be bearable. Carelessness you can dismiss. What you cannot dismiss is precision in the service of nothing. The Pit is extremely precise.

Each stone at level k is inscribed with exactly $2k - 1$ words. Every word is a different synonym for *absence*. Void. Null. Naught. Vacuity. Oblivion. The-space-where-something-was. The Pit has been compiling this thesaurus for longer than language has existed, which raises questions about method that Soren has decided not to look into.

Level 1: one word. Level 2: three words. Level 3: five. Level 4: seven. The odd numbers, forever, like the Pit is trying to describe itself and keeps needing more words and never quite gets there.

Soren notices, because he cannot stop noticing, that the total number of words through level n is always a perfect square. Always n squared. A square is the most self-contained shape there is. A number times itself. It refers to nothing outside itself. He finds this, if not comforting, then at least consistent with everything else.

$$1 + 3 + 5 + \dots + (2n-1) = n^2$$

Prove this by mathematical induction.

“Soren found this result and sat with it for a while. He did not feel better. He kept falling.”

(c) The Silence Wins [9 Marks]

Here is where the Pit stops being merely sad and becomes something harder to name.

Between levels, there is silence. Not the comfortable kind between two people who know each other well enough to say nothing. Not the quiet of a library, which is a silence that knows books exist nearby. This is the original silence. The one that was there before the universe started making noise, and will be there again after it stops.

Soren measures the silence between level k and level $k+1$. It has magnitude 2^k . It doubles with every descent. Relentless in the way that only things without feelings can be relentless, without effort, without awareness, just growing because that is what it does.

The total words on stones through level n : a clean n^2 . Polynomial. Predictable. The kind of growth that starts off confident and then quietly watches the exponential pull ahead, and then lose sight of it entirely, and then forget what it was even chasing.

Soren looks at both numbers for a long time. Then he writes, in the air that has been absorbing his writing this whole time without once asking why:

$$2^n \geq n^2 \quad \text{for all integers } n \geq 4$$

Prove this by mathematical induction.

“The silence grows faster than the words for silence. The void outpaces its own vocabulary.”

“Soren thought about this for the rest of the fall, which is to say: forever. He did not find it comforting. He found it true. In his experience, those two things had always been different.”